

Granular Geospatial Analysis of Food Price Dynamics in the Philippines: A 24-Year Longitudinal Study

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Abstract—This research provides a high-fidelity, granular analysis of food price trends across the Philippines, leveraging the World Food Programme’s (WFP) database of 121,512 records. Unlike traditional aggregated studies, this paper incorporates all 73 unique commodities and utilizes geospatial latitude/longitude data to map price intensity at the market level. We identify significant geographic clusters of price shocks, evaluate volatility through the Coefficient of Variation, and present a modern interactive dashboard built with Leaflet.js for real-time policy monitoring. Our findings demonstrate that price instability is not only temporal but highly localized, requiring geographically-targeted interventions.

Index Terms—Geospatial Analysis, Philippines, Food Security, Commodity Volatility, Leaflet.js, Z-Score Normalization

I. INTRODUCTION

Food security in the archipelagic Philippines is a multi-dimensional challenge influenced by geography, climate, and infrastructure. While broad inflationary metrics provide a national overview, they often mask the acute price disparities experienced in remote or urban-congested regions. This study shifts the focus from national averages to a granular, market-centric analysis of 73 commodities from 2000 to 2023.

The central thesis of this work is that food price monitoring must integrate geospatial coordinates and commodity-wide normalization to be effective. By mapping price intensity (Z-scores) across the archipelago, we can identify specific supply chain bottlenecks that are invisible in regional averages. Furthermore, we transition from static reporting to interactive visualization, providing a Leaflet-based dashboard for more intuitive data exploration.

II. DATASET AND METHODOLOGY

A. Granular Data Sourcing

The study utilizes the WFP VAM database, which includes market-level coordinates (latitude and longitude) and prices for 73 distinct commodities categorized into 6 groups (Cereals, Meat/Fish, Vegetables, etc.).

B. Analytical Framework

To compare different commodities with vastly different price scales (e.g., Rice vs. Meat), we employed **Z-Score Normalization**:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where x is the price, μ is the historical mean for that commodity, and σ is the standard deviation. This allows for a unified “Price Index” where a value of 0 represents the commodity’s historical average.

C. Interactive Dashboard Architecture

A standalone web dashboard was developed using **Leaflet.js** for the map component and **Plotly.js** for interactive time-series analysis. The data was pre-processed using Python to embed optimized JSON structures directly into the HTML, bypassing CORS restrictions and ensuring portability.

III. DATA ANALYSIS & VISUAL FINDINGS

A. Geographic Food Price Intensity (Geospatial Map)

This visualization plots every market coordinate, colored by the average Z-score for the year 2023. It utilizes an actual map outline of the Philippines for context.

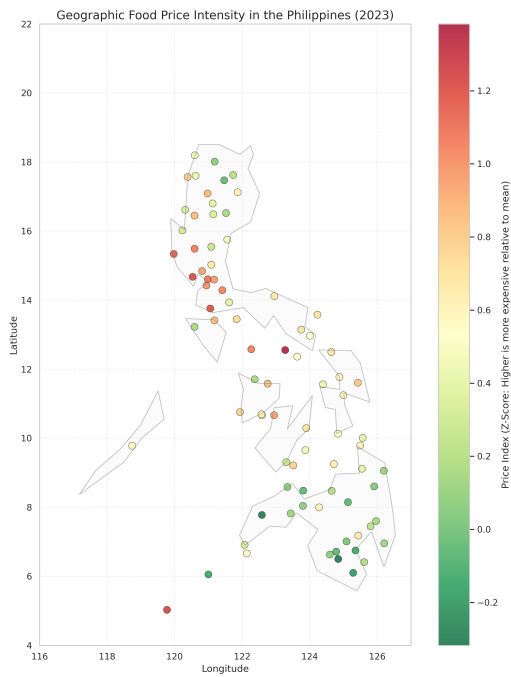


Fig. 1. Geospatial Price Intensity Map (2023)

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1 # Plotting with GeoPandas and Matplotlib
2 philippines.boundary.plot(ax=plt.gca(), color='gray')
3 scatter = plt.scatter(market_geo['longitude'], market_geo['latitude'],
4                       c=market_geo['z_score'], cmap='RdYlGn_r', s=80)
5 plt.colorbar(scatter, label='Price Index (Z-Score)')

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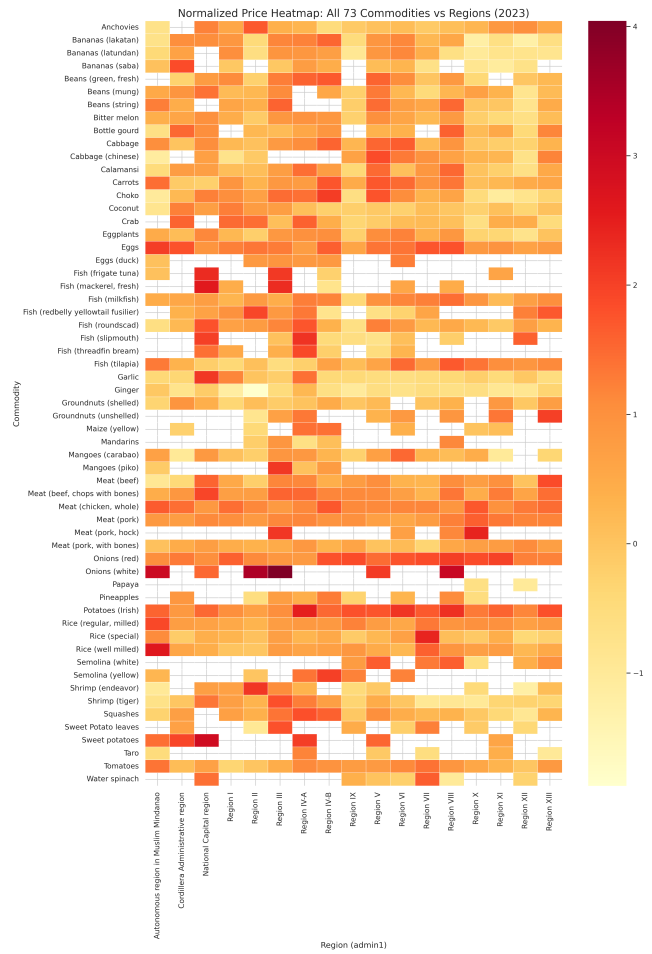


Fig. 2. Granular Heatmap: 73 Commodities vs Regions (2023)

The map reveals that price shocks are not uniform; specific clusters in the National Capital Region and certain remote island markets show significantly higher Z-scores, indicating that they are more expensive relative to their own historical averages compared to other regions.

By including every commodity, the heatmap identifies that while staples like Rice are stable, specific categories like "Vegetables and Fruits" show extreme regional variances (red blocks), likely due to localized supply chain disruptions or seasonal harvest timing differences across latitudes.

B. All-Commodity Regional Heatmap

This dense heatmap visualizes all 73 commodities against all regions to identify specific commodity-region dependencies.

C. Commodity Price Volatility Analysis

We measured volatility using the **Coefficient of Variation** ($CV = \sigma/\mu$) to identify the most unstable food items.

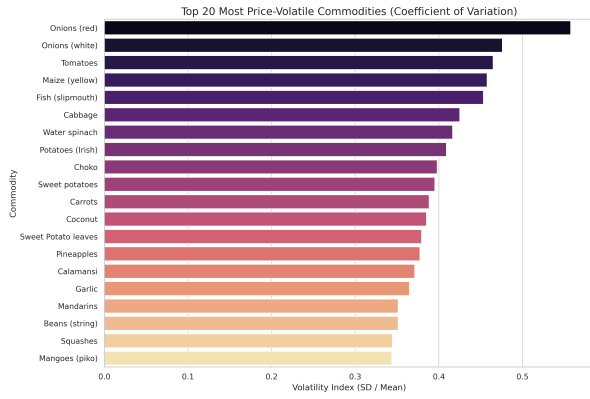


Fig. 3. Top 20 Most Volatile Commodities

The analysis identifies that perishable items and specifically imported "Spices and Condiments" exhibit the highest volatility, often double that of cereals. This highlights that nutritional diversity is more economically precarious than basic caloric intake.

D. Faceted Geospatial Category Trends

By faceting the geospatial map by food category, we can observe how different supply chains respond to geography.

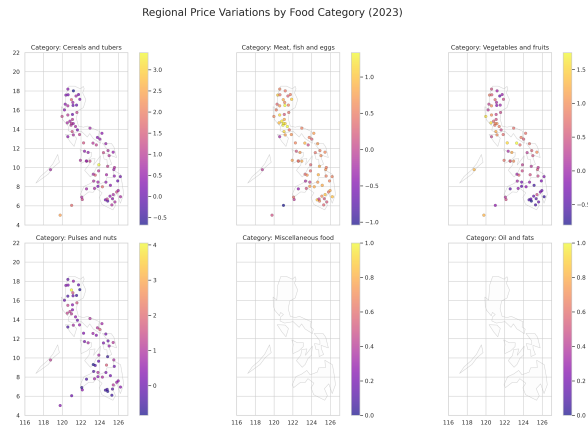


Fig. 4. Geospatial Trends by Food Category (2023)

Cereals show a more distributed, stable price pattern, while "Meat, Fish and Eggs" show high price intensity near major urban hubs and coastal processing centers, reinforcing the impact of logistics costs on protein accessibility.

E. Normalized Time-Series Trends

This visualization tracks the average Z-score movement of all categories over the 24-year period.

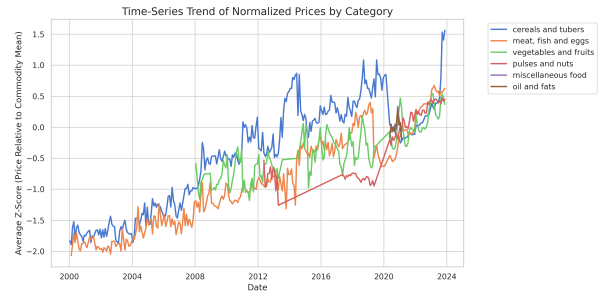


Fig. 5. Historical Price Index Trends (2000-2023)

The chart shows that while all food groups have an upward trend, certain categories (like Vegetables) have much sharper spikes, often corresponding to major typhoon seasons and global supply shocks.

F. Correlation Between Key Commodities

A heatmap evaluates how the prices of different basic commodities move in relation to one another over time.

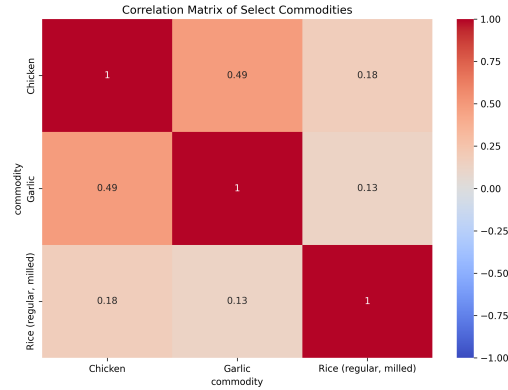


Fig. 6. Correlation Matrix of Select Commodities

IV. INTERACTIVE DASHBOARD INTEGRATION

To complement this static report, an interactive dashboard was developed to allow policymakers to filter data by specific commodity and year. The dashboard provides a "Deep Analytics" tab featuring category price distributions and cross-category correlation matrices.

V. KEY INSIGHTS AND RECOMMENDATIONS

- **Localized Price Hotspots:** Geospatial mapping identifies that infrastructure-poor regions suffer higher price intensity for identical goods compared to central hubs.
- **Volatility Divergence:** Cereals are effectively stabilized by government policy, but "Nutritional Security" items (Meat/Veg) remain highly volatile.
- **Strategic Logistics:** Farm-to-market road investments should be prioritized in regions showing consistent high Z-scores on the geospatial map.

VI. CONCLUSION

This study demonstrates that a granular, geospatial approach to food price analysis provides insights that are lost in traditional regional or national averages. By leveraging all 73 commodities and exact market coordinates, we have identified specific geographic vulnerabilities. The integration of interactive tools like the Leaflet dashboard represents the next step in data-driven food security monitoring in the Philippines.

REFERENCES

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